

KARTHIK BHARADWAJ

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

EDUCATION

Northeastern University - Boston, MA | *Master of Science, Computer Science* | **GPA: 4.0** Sept 2025 - Expected Dec 2027
Relevant Coursework: Programming Design Paradigms, Database Management Systems, Algorithms, Foundations of AI, Building Scalable Distributed Systems (Fall 2026)

R V College of Engineering - Bangalore, India | *Bachelor of Engineering, Computer Science and Engineering* 2023

PROFESSIONAL EXPERIENCE

Digital Associate - Software Engineer | *Digitide Solutions Ltd (Client: AB-InBev)* - Bangalore, India Sept 2024 - June 2025

- Engineered a modular SaaS platform serving 1,000+ daily users with report generation, admin controls, planogram extraction, and bulk update/delete workflows using Angular, Python, Docker, and Azure.
- Designed and integrated real-time operational dashboards consuming sensor and transactional data streams, enabling proactive alerting and data-driven decision-making across warehouse operations.

Software Engineer - Full Stack | *Zevvo* - Bangalore, India Jan 2024 - July 2024

- Built a full admin portal and customer-facing pages from scratch using React, Node.js, and Express, delivering the product from zero to production in under 6 months.
- Automated CI/CD pipelines via GitHub Actions deploying Dockerized services to AWS ECS with auto-scaling, increasing deployment frequency by 40% and enabling same-day hotfixes for production issues.

Software Development Intern | *Crestron Electronics* - Bangalore, India March 2023 - July 2023

- Built mobile-responsive React components with Google Tag Manager integration, driving behavioral analytics that informed the product feature roadmap.
- Developed and deployed .NET Core REST APIs with optimized SQL queries, cutting average response time by 28%, directly improving page load speeds for end users across the platform.

TECHNICAL SKILLS

- Languages:** TypeScript, JavaScript, Python, Java, C#, C++, SQL, HTML/CSS
- Frameworks and Frontend:** React.js, Next.js, Node.js, Express.js, Angular, FastAPI, .NET Core, Tailwind CSS, REST APIs
- Cloud and Infrastructure:** AWS (ECS, S3, Lambda, EC2, ECR), Azure (Container Registry, Blob Storage), Firebase, Docker, GitHub Actions, CI/CD
- Data and Databases:** MongoDB, PostgreSQL, MySQL, Neo4j, MSSQL, SQLite, Firebase Realtime DB
- Engineering Practices:** System Design (HLD/LLD), Agile/Scrum, TDD, OOP, Responsive Design, Code Reviews, Git, Jest

CERTIFICATIONS

Data Structures, Coursera (December 2020) | Python Data Structures, Coursera (December 2021)

ACADEMIC PROJECTS

ParetoFolio: Portfolio Optimization Engine | [GitHub](#) - TypeScript, React 19, FastAPI, Python, Pydantic v2 Spring 2026

- Built a responsive React 19 frontend with interactive data visualization and a typed FastAPI backend, serving a RESTful API for multi-objective portfolio optimization across 75 US equities.
- Implemented NSGA-II from scratch with SBX crossover, polynomial mutation, tournament selection, and crowding distance, producing Pareto-optimal portfolios with an interactive efficient frontier and Sharpe-ranked table.

VyasaGraph: Knowledge Graph & RAG Chatbot | [GitHub](#) - Python, FastAPI, React 19, TypeScript, Neo4j, ChromaDB, spaCy, Ollama 2026

- Built a knowledge graph over a 1.8M-word corpus with 228 characters and 315 relationships sourced from three independent pipelines: spaCy dependency-parsed attestations, a structured CSV dataset, and Wikidata SPARQL queries, with human review and deduplication across all sources.
- Implemented a graph-augmented RAG pipeline combining Neo4j Cypher queries with ChromaDB semantic search (960 embedded chunks via sentence-transformers) to ground LLM answers in both structured facts and source text, served through a FastAPI streaming backend and a React chat interface with real-time citation display.

LEADERSHIP AND ACTIVITIES

National Service Scheme (NSS) | January 2020 - August 2023 - Organized community projects including school renovations, awareness campaigns, and rural outreach programs for societal development.

Youth for Seva | January 2022 - December 2022 - Part of a four-member team assigned to a rural school for teaching and innovation. Organized an exhibition encouraging students to apply concepts learned.